

LMM Report – HGS_MAX

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Imputations: 10

Random slope: FALSE | Interaction: FALSE

Exposures: 7 | Covariate sets: 2

Model Configuration

Outcome column: HGS_MAX
Outcome transform: identity
Estimation method: REML = FALSE (Maximum Likelihood)
Random slope: FALSE
Interaction (exposure × time): FALSE

Exposures:

	exposure	exposure_type	ref_level
1	dairy_100g	linear	<NA>
2	fermented_100g	linear	<NA>
3	nonfermented_100g	linear	<NA>
4	highfat_100g	linear	<NA>
5	lowfat_100g	linear	<NA>
6	dairy_quartile_baseline	categorical	Q1
7	dairy_guidelines_port	categorical	< 2 servings/day

Covariate sets:

main:

age_at_baseline_scaled, BMI_category, education_level, smoking_status, pa_levels_tertile_f1, diabetes_status, sumtot1_scaled

other_PA:

age_at_baseline_scaled, BMI_category, education_level, smoking_status, pa_levels_who_f1, diabetes_status, sumtot1_scaled

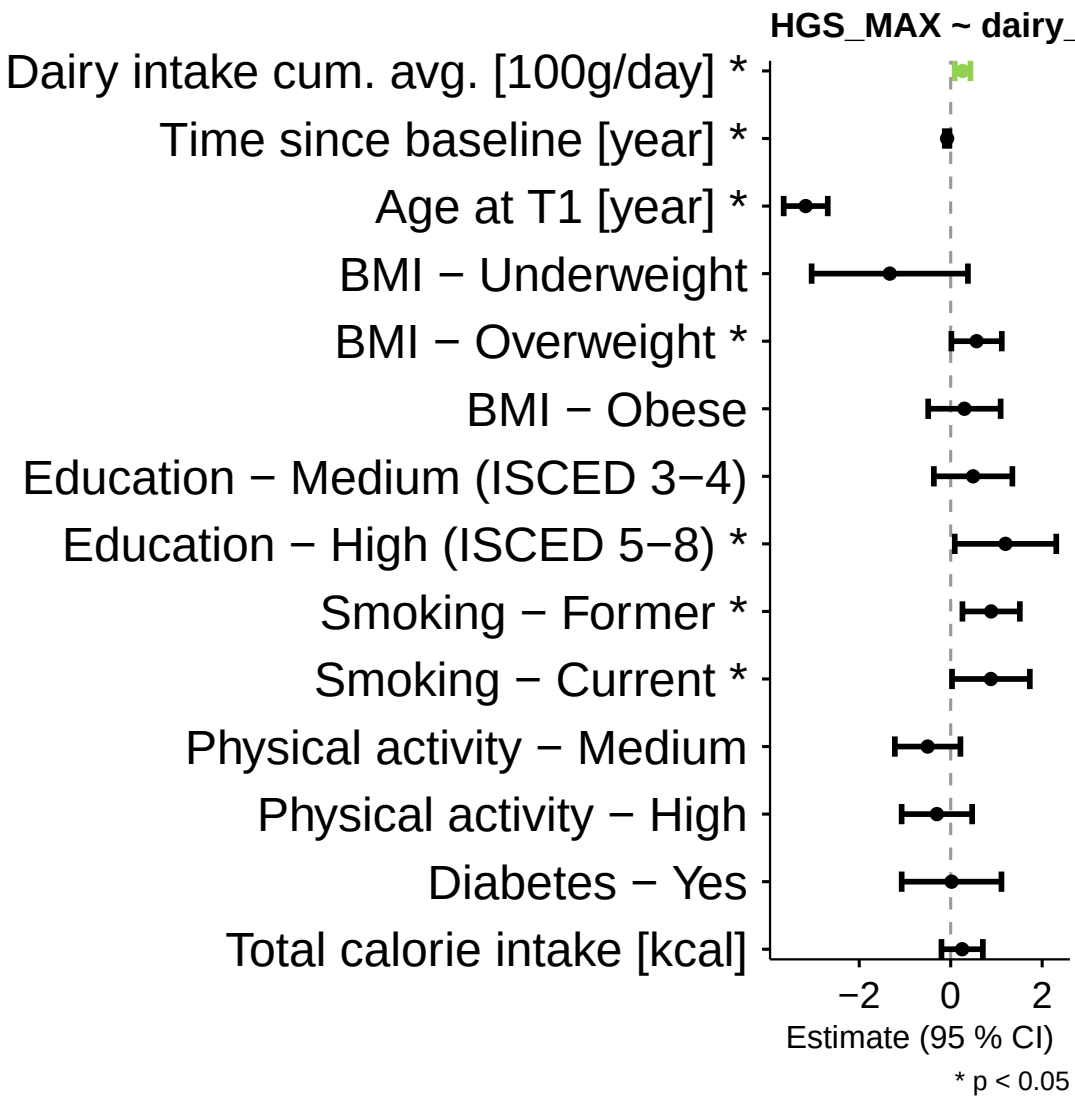
Covariate set: main

age_at_baseline_scaled · BMI_category · education_level · smoking_status · pa_levels_tertile_f1 · diabetes_status · sumtot1_scaled

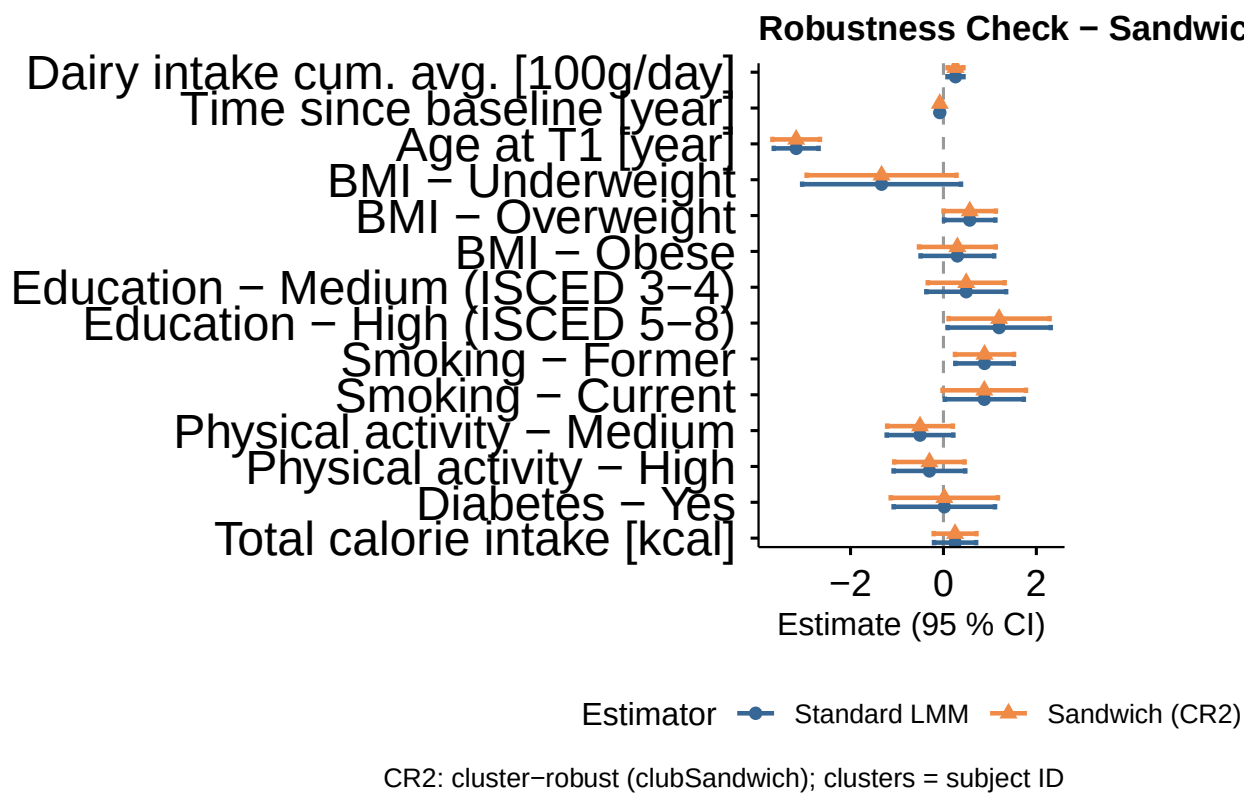
Formula – HGS_MAX ~ dairy_100g (linear) | main

HGS_MAX ~ dairy_100g + time_since_baseline + age_at_baseline_scaled +
BMI_category + education_level + smoking_status + pa_levels_tertile_f1 +
diabetes_status + sumtot1_scaled + (1 | pt)

Imputations pooled: 10



term	estimate	std_error	statistic	df	p_value	conf_low	conf_high
(Intercept)	23.497	0.537	43.735	2134.8	<0.001	22.444	24.551
dairy_100g	0.259	0.087	2.966	2164.5	0.003	0.088	0.431
time_since_baseline	-0.079	0.026	-3.042	2168.0	0.002	-0.129	-0.028
age_at_baseline_scaled	-3.170	0.246	-12.903	2194.9	<0.001	-3.652	-2.688
BMI_categoryUnderweight	-1.332	0.871	-1.529	2198.7	0.127	-3.041	0.377
BMI_categoryOverweight	0.566	0.281	2.015	2196.3	0.044	0.015	1.117
BMI_categoryObese	0.300	0.404	0.742	2196.6	0.458	-0.492	1.092
education_levelMedium (ISCED 3–4)	0.490	0.438	1.121	2188.0	0.263	-0.368	1.348
education_levelHigh (ISCED 5–8)	1.198	0.566	2.117	2192.0	0.034	0.088	2.308
smoking_statusFormer	0.884	0.320	2.762	2195.3	0.006	0.256	1.512
smoking_statusCurrent	0.879	0.434	2.028	2165.6	0.043	0.029	1.730
pa_levels_tertile_f1Medium	-0.504	0.366	-1.379	1930.9	0.168	-1.221	0.213
pa_levels_tertile_f1High	-0.302	0.393	-0.767	1979.4	0.443	-1.073	0.469
diabetes_statusDiabetes	0.019	0.556	0.035	2143.6	0.972	-1.072	1.110
sumtot1_scaled	0.250	0.231	1.086	1394.5	0.278	-0.202	0.703



term	estimate	SE_std	p_std	SE_sandwich	p_sandwich	delta_SE
(Intercept)	23.497	0.537	<0.001	0.508	<0.001	-0.029
dairy_100g	0.259	0.087	0.003	0.086	0.003	-0.001
time_since_baseline	-0.079	0.026	0.002	0.027	0.004	0.001
age_at_baseline_scaled	-3.170	0.246	<0.001	0.260	<0.001	0.014
BMI_categoryUnderweight	-1.332	0.871	0.127	0.822	0.105	-0.049
BMI_categoryOverweight	0.566	0.281	0.044	0.287	0.049	0.006
BMI_categoryObese	0.300	0.404	0.458	0.421	0.476	0.017
education_levelMedium (ISCED 3-4)	0.490	0.438	0.263	0.422	0.245	-0.016
education_levelHigh (ISCED 5-8)	1.198	0.566	0.034	0.554	0.031	-0.012
smoking_statusFormer	0.884	0.320	0.006	0.324	0.006	0.004
smoking_statusCurrent	0.879	0.434	0.043	0.458	0.055	0.024
pa_levels_tertile_f1Medium	-0.504	0.366	0.168	0.360	0.162	-0.006
pa_levels_tertile_f1High	-0.302	0.393	0.443	0.386	0.435	-0.007
diabetes_statusDiabetes	0.019	0.556	0.972	0.588	0.974	0.032
sumtot1_scaled	0.250	0.231	0.278	0.234	0.285	0.003

Variance Components – HGS_MAX ~ dairy_100g (linear) | main

```
.. Full VarCorr output .....
Groups   Name          Variance Std.Dev.
pt       (Intercept) 14.535    3.8125
Residual              13.524    3.6774

.. Summary .....
Random intercept variance: 14.53531 (SD = 3.81252)
Random slope: not included in this model
Residual variance:         13.52362 (SD = 3.67745)
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Effect Size (R²) – HGS_MAX ~ dairy_100g (linear) | main

Marginal R² (fixed effects only): 0.1514

Conditional R² (fixed + random effects): 0.591

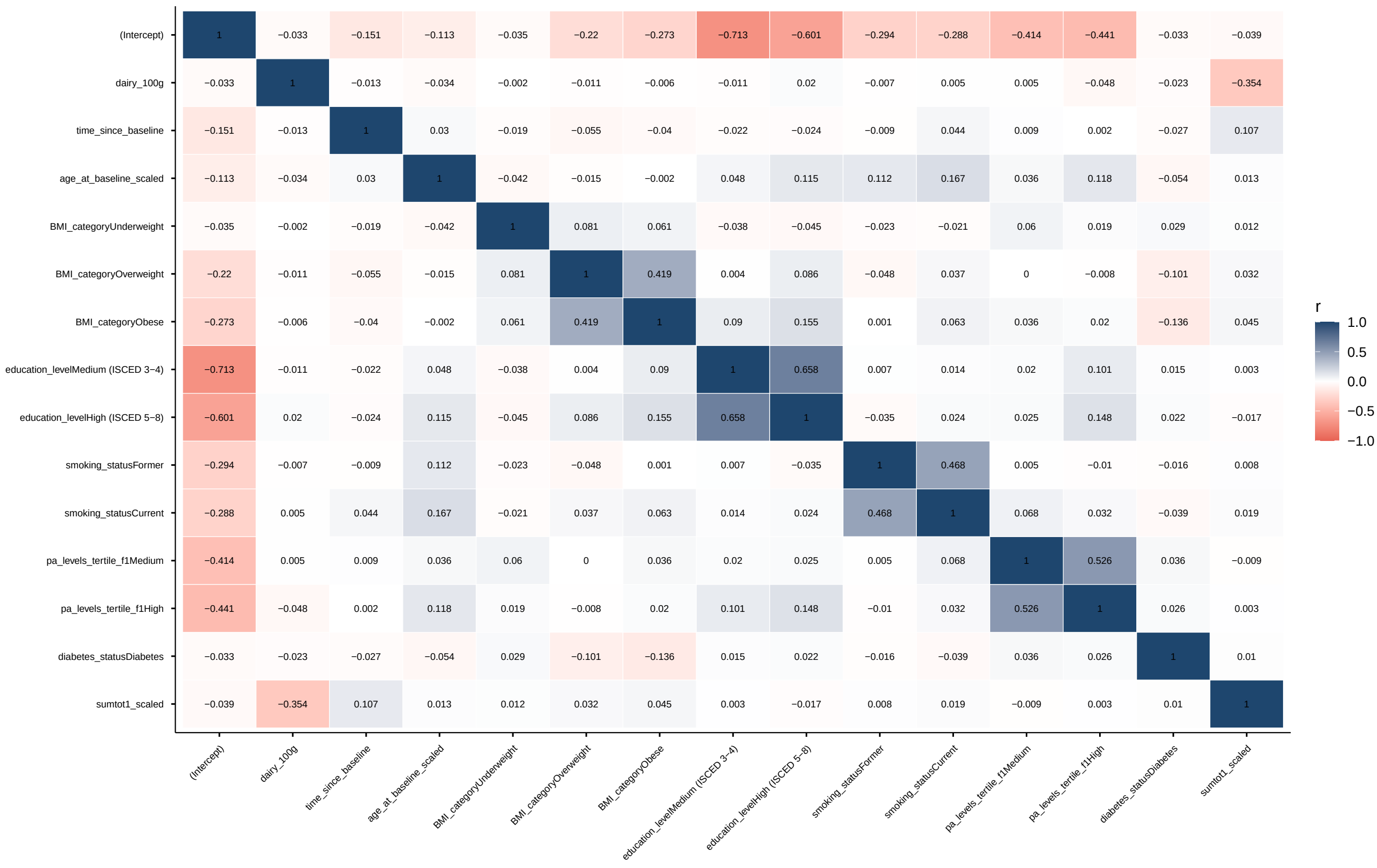
Method: Nakagawa & Schielzeth (2013), Johnson (2014)

VIF – HGS_MAX ~ dairy_100g (linear) | main

VIF > 5 flagged with '!'

term	GVIF	Df	$GVIF^{1/(2 \cdot Df)}$	adjusted_vif	flag
dairy_100g	1.152084	1	1.073352	1.073	
time_since_baseline	1.022503	1	1.011189	1.011	
age_at_baseline_scaled	1.065524	1	1.032242	1.032	
BMI_category	1.086094	3	1.013860	1.014	
education_level	1.085183	2	1.020647	1.021	
smoking_status	1.055759	2	1.013657	1.014	
pa_levels_tertile_f1	1.056175	2	1.013757	1.014	
diabetes_status	1.033266	1	1.016497	1.016	
sumtot1_scaled	1.162041	1	1.077980	1.078	

Fixed-Effect Correlations – HGS_MAX ~ dairy_100g (linear) | main



Influential Points – HGS_MAX ~ dairy_100g (linear) | main

Influential observations (top 1% Cook's D > 0.24762): 23

Index	Cook's D
index	cooks_d
13	0.29529
384	0.26101
560	0.28372
629	0.43662
784	0.27593
1051	0.37296
1121	0.38670
1292	0.24770
1333	0.49069
1391	0.26850
1450	0.33245
1471	0.41811
1533	0.31829
1610	0.38463
1630	0.31642
1694	0.73527
1709	0.41042
1750	0.35169
1952	0.33380
1975	0.30842
1993	0.65572
2044	0.34008
2166	0.32492